Prayas JEE (2024)

Trigonometric Equations

Revision Practice Sheet-1

JEE

Single Correct Type Questions (1-15)

1. If $\alpha < \beta < \gamma$ and $\sin\gamma\cos\alpha = 1$, where $\alpha, \gamma \in [\pi, 2\pi]$, then the least integral value of $f(x) = |x - \alpha| + |x - \beta| + |x - \gamma|$ is

(1) 0 (2) 1
(3) 2 (4) 3

2. The number of solutions of the equation $5^{\frac{1}{2}} + 5^{\frac{1}{2}+\log_5(\sin x)} = 15^{\frac{1}{2}+\log_{15}\cos x}$ for $x \in [0, 100\pi]$ is

(1) 50
(2) 100
(3) 200
(4) 400

3. Let $x + \sin y = 2009$ and $x + 2009\cos y = 2008$, where $y \in \left[0, \frac{\pi}{2}\right]$. Then the value of $[x + y]$, where $[\cdot]$ represents the greatest integer function.

(1) 2008
(2) 2009
(3) 2100
(4) 2010

4. The number of solutions of the equation $\sin\frac{5x}{2} - \sin\frac{x}{2} = 2$, $x \in [0, 2\pi]$, is

(1) 0 (2) 1
(3) 2 (4) 3

5. If the equation $k\sin x + \sqrt{k-2}\cos x + (\tan\alpha + \cot\alpha) = 0$, $0 < \alpha < \pi/2$, possesses real solution, then k belongs to

(1) $(-\infty, -3] \cup [2, \infty)$
(2) $[-3, 2]$
(3) $[0, 2]$
(4) R

6. If the equation $x^2 + 12 + 3\sin(a + bx) + 6x = 0$ has at least one real solution where $a, b \in [0, 2\pi]$, then the value of $(a - 3b)$ is

(1) $2n\pi$ (2) $(2n + 1)\pi$
(3) $(4n-1)\frac{\pi}{2}$ (4) $(4n+1)\frac{\pi}{2}$

7. Which of the following statements is correct about the solution set of the system of equations $x + y = 2\pi/3$ and $\cos x + \cos y = 3/2$, where x and y are real?

(1) Solutions is: $n\pi + (-1)^n \sin^{-1}\frac{3}{2} + \frac{\pi}{6}$
(2) Solution is: $2n\pi \pm \cos^{-1}\frac{3}{2} - \frac{\pi}{6}$
(3) There is no solution
(4) None of these

8. The number of distinct real roots of the equation $\sin(\pi x) = x^2 - x + \frac{5}{4}$ is

(1) 0 (2) 1
(3) 2 (4) 4

9. The set of all possible values of p for which $\sqrt{p}\sin x - 2\cos x = \sqrt{2} + \sqrt{2-p}$ has solutions are

(1) $p > 0$ (2) $p \le 3$
(3) $1 \le p \le 2$ (4) $\sqrt{5} - 1 \le p \le 2$

10. If $\sin x + \cos x = y^2 - y + a$ is not satisfied by any real value of x for any y, then $a \in$

(1) $\left(-\infty, -\sqrt{2} + \frac{1}{4}\right)$ (2) $\left(\sqrt{2} + \frac{1}{4}, \infty\right)$
(3) $\left(\sqrt{2} - \frac{1}{4}, \infty\right)$ (4) $\left(-\infty, \sqrt{2} - \frac{1}{4}\right)$

11. If $|\operatorname{cosec} x| = \frac{5\pi}{4} - \left|\frac{x}{2}\right|, \forall x \in (-2\pi, 2\pi)$, then the number of solutions is

(1) 8 (2) 6
(3) 4 (4) 2

12. The number of solutions of the equation $|\sin x| = |\cos 3x|$, where $x \in [-2\pi, 2\pi]$, is
(1) 32 (2) 28
(3) 24 (4) 30

13. If $6|\sin x| = x$ when $x \in [0, 2\pi]$, then number of solutions is
(1) 0 (2) 2
(3) 4 (4) 5

14. If $2\sin^2\left(x-\frac{\pi}{3}\right)-5\sin\left(x-\frac{\pi}{3}\right)+2<0$, then x belongs to
(1) $\left(\frac{(12n-5)\pi}{6},\frac{(4n+1)\pi}{2}\right), n \in Z$
(2) $\left(\frac{(6n-7)\pi}{6},\frac{(2n+1)\pi}{2}\right), n \in Z$
(3) $\left(\frac{(4n+1)\pi}{6}, n\pi\right), n \in Z$
(4) $\left((4n+1)\frac{\pi}{2},(12n+7)\frac{\pi}{6}\right), n \in Z$

15. If $A=\{x \mid 2\sin^2 x+3\sin x-2>0\}$ and $B=\{x \mid 2\cos^2 x - 3\cos x - 2 > 0\}$, then for $A \cap B$.
(1) $x \in \left(\frac{\pi}{3},\frac{\pi}{2}\right)$ (2) $x \in \left(\frac{\pi}{6},\frac{\pi}{2}\right)$
(3) $x \in \left(\frac{\pi}{6},\frac{5\pi}{6}\right)$ (4) $x \in \left(\frac{2\pi}{3},\frac{5\pi}{6}\right)$

Multiple Correct Answers Type (16-17)

16. Which of the following are the solutions of equation $2\sin 11x + \cos 3x + \sqrt{3}\sin 3x = 0$?
(1) $x=\frac{n\pi}{7}-\frac{\pi}{84}, n \in Z$
(2) $x=\frac{n\pi}{4}+\frac{7\pi}{48}, n \in Z$
(3) $x=\frac{n\pi}{7}-\frac{\pi}{63}, n \in Z$
(4) $x=\frac{n\pi}{4}+\frac{\pi}{24}, n \in Z$

17. If $3^{\left(\frac{1}{2}+\log_3(\cos x+\sin x)\right)} - 2^{\log_2(\cos x-\sin x)} = \sqrt{2}$ then $x =$ (Here, $n \in Z$)
(1) $2n\pi+\frac{3\pi}{4}$
(2) $n\pi+(-1)^n\frac{\pi}{12}$
(3) $2n\pi+\frac{\pi}{12}$
(4) $n\pi+(-1)^n\frac{\pi}{3}$

Linked Comprehension Type
For Question (18-19)

Consider the equation $2^{\sin x}+2^{\cos x}=2^{1-\frac{1}{\sqrt{2}}}$.

18. The equation is equivalent to the equation
(1) $\sin x + \cos x = 0$
(2) $\sin x = \cos x$
(3) $\cos 2x = 0$
(4) $\sin x + \cos x = -\sqrt{2}$

19. The number of solutions of the equation in the interval $[0, 100\pi]$ is
(1) 50 (2) 100
(3) 200 (4) 400

Matrix Match Type Question (20-22)

20. Match the trigonometric equations given under List I with their respective general solutions given under List II.

	List I		List II
(A)	$\log_{12}(\tan\theta + 3\cot\theta) = \frac{1}{2}$	**(i)**	$2n\pi-\frac{\pi}{6}$, n ∈ Z
(B)	$2\cos^4 x + \cos x - 2\cos x \sin^2 x - 3\sin^2 x + 1 = 0$	**(ii)**	$n\pi+\frac{\pi}{3}$, n ∈ Z
(C)	$\sin^2 2x - \cos^2 8x = \frac{1}{2}\cos 10x$	**(iii)**	$n\pi\pm\frac{\pi}{4}$, , n ∈ Z
(D)	$2\cot^2 x + 2\sqrt{3}\cot x + 4\operatorname{cosec} x + 8 = 0$	**(iv)**	$\frac{n\pi}{3}\pm\frac{\pi}{9}$, n ∈ Z

(1) (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)
(2) (A)-(ii), (B)-(iii), (C)-(iv), (D)-(i)
(3) (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii)
(4) (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)

21. Match the equations given under List I with their respective numbers of solutions given under List II.

	List I	List II	
(A)	$\sqrt{\sin x} = \cos x;\ x \in [0, 2\pi]$	**(i)**	0
(B)	$\tan x = \sec x;\ x \in [0, 2\pi]$	**(ii)**	1
(C)	$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+\sin\theta & 1 \\ 1 & 1 & 1+\cot\theta \end{vmatrix} = 0;\ \theta \in [0, 4\pi]$	**(iii)**	2
(D)	$\sin^3 x + \sin^2 x + \sin x - \sin x \sin 2x - \sin 2x - 2\cos x = 0;\ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	**(iv)**	4

(1) (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)
(2) (A)-(ii), (B)-(iii), (C)-(iv), (D)-(i)
(3) (A)-(ii), (B)-(i), (C)-(iv), (D)-(ii)
(4) (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)

22. Match the statement of Column I with the value of Column II.

	Column I	Column II	
(A)	The number of solutions of the equation $\lvert\tan 2x\rvert = \sin x;\ x \in [0, \pi]$	**(i)**	1
(B)	The value of $4\tan\frac{\pi}{16} - 4\tan^3\frac{\pi}{16} + 6\tan^2\frac{\pi}{16} - \tan^4\frac{\pi}{16} + 1$	**(ii)**	4
(C)	If the equation $\tan(p\cot x) = \cot p(\tan x)$ has a solution in $(0, x) - \left\{\frac{\pi}{2}\right\}$, then $\frac{4}{\pi}P_{max}$ is	**(iii)**	3
(iv)	The value of $\frac{2x}{\pi}$ in $[0, 2\pi]$ if $5^{\cos^2 2x + 2\sin^2 x} + 5^{2\cos^2 x + \sin^2 2x} = 126$ has a solution	**(iv)**	2

(1) (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)
(2) (A)-(ii), (B)-(iv), (C)-(i), (D)-(i),(iii)
(3) (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii)
(4) (A)-(iv), (B)-(i), (C)-(ii), (D)-(i),(iii)

<u>Numerical Value Type Question (23-25)</u>

23. Consider the following inequality:

$$2^{\sin^2 u + 3\cos^2 v} \times 3^{\sin^2 w + \cos^2 x} \times 5^{\cos^2 y} \geq 720,$$

where $u, v, w, x, y \in [1, 11]$. The number of ordered 5-tuples (u, v, w, x, y) is ________.

24. The number of solutions of the equation $\sin x \cdot \sin 2x \cdot \sin 3x = 1$, where $x \in [0, 100\pi]$, is _________.

25. The number of ordered pairs (α, β), where $\alpha, \beta \in [0, 2\pi]$ satisfying $\log_{2\sec x}(\beta^2 - 6\beta + 10) = \log_3\lvert\cos \alpha\rvert$ is

Hints and Solutions

Single Correct Answer Type

1. (3)
2. (1)
3. (2)
4. (1)
5. (1)
6. (3)
7. (3)
8. (2)
9. (4)
10. (2)
11. (1)
12. (3)
13. (3)
14. (4)
15. (4)

Multiple Correct Answers Type

16. (1, 2)
17. (1, 3)

Linked Comprehension Type

18. (4)
19. (1)

Matrix Match Type Question

20. (2)
21. (3)
22. (2)

Numerical Value Type

23. (432)
24. (0)
25. (2)